Ethical Considerations of AI-Generated Evidence in Digital Forensics

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I am majoring in Cybersecurity with a career goal of working in digital forensics. As AI is being used more and more in forensic investigations, it is important to understand how it is being used to build evidence. This paper looks at the ethical issues that come up when AI-generated evidence is used in a court of law.

AI use in digital forensics helps in many ways. It can automate data processing, as there can be massive amounts of evidence to collect from phones, IoT devices, and computers, for example. AI can be used for pattern recognition that possibly investigators may miss as well as it can process data much quicker than humans. AI can enhance images or videos to improve clarity or used for facial recognition. These are all useful tools, but the issue is whether any of the evidence holds up in court. Courts are skeptical of "black box" AI in digital forensics because if no one can explain how AI reached its conclusion, the evidence becomes hard to trust. (Solanke, 2022). Judges and juries do not have the technical background to understand how AI works and this lack of transparency makes evidence questionable. This creates a barrier for the defense to challenge the AI evidence they do not understand. There are also concerns about defendants' rights when AI is involved. The defense needs to be able to access the evidence, understand how AI reached its conclusions, and have a say in how digital evidence is processed (Stoykova, 2024). If the evidence is not transparent, it could violate someone's right to a fair trial and could damage trust in the whole system. It could also create a justice system where outcomes depend on the defendants having the resources to challenge AI evidence.

The United States Army's Ethical Decision-Making Framework follows a structured stepped approach to solve ethical challenges. I am applying this framework to the AI evidence admissibility problem. The courts are rejecting or questioning AI-generated evidence because AI

systems lack transparency, which makes it difficult to verify reliability and put together steps for accountability when AI makes errors.

Looking at this from the Christian worldview, the main concern is whether AI evidence respects human dignity and upholds justice. Micah 6:8 teaches acting with fairness, righteousness, and integrity. AI evidence must be transparent and easily explained so truth can be established. Defendants have the right to a fair trial and cannot be convicted on evidence that is not understood or that cannot be challenged. The people that build and are using these AI tools have a moral obligation to make sure the evidence is honest and can be verified. From a Christian perspective, the “black box” evidence that no one can explain goes against these principles.

From a professional standpoint, the ACM Code of Ethics addresses this issue. This view would require AI tools be developed so the courts can verify how AI reached its conclusions to prevent wrongful convictions or guilty parties going free. Developers must be transparent about limitations of their AI models and courts would need to understand how it works before accepting their evidence. When mistakes are made, those responsible should be held accountable, whether the developer, forensic lab, or investigating agency. This includes keeping record of how the AI system made its decisions and making sure performance data is available if it goes to court. The AI model cannot have bias because that destroys any chance of a fair trial. If an AI forensic tool cannot meet basic standards for transparency and accountability, it should not be admissible in court (ACM, 2018).

The IEEE Code of Ethics has a similar perspective but focuses more on human rights and well-being. Their position is that everyone has a right to a fair trial, which means evidence must be understandable and is something the defense can challenge. If the court, defendant, and the

attorneys cannot understand how AI came to its conclusion, that is a problem under the IEEE standards. When AI makes a mistake that affects a criminal case, someone must be held accountable. IEEE standards state that human well-being and fairness must be prioritized over technology. AI should not be used in forensics for convenience if it cannot meet basic standards of explainability and reliability (IEEE, 2019).

AI systems are not neutral and there are biases that can happen when AI is used in forensic work. AI models would typically be trained on past cases which may cause biases. AI may make assumptions based on demographics by flagging innocent behavior as criminal or criminal behavior as innocent. AI may use patterns that would confirm what investigators suspect causing the AI model or even the investigators to miss alternative explanations. AI may have biases regarding evidence types. The model may be trained well in some areas, but weaker in others. This may cause the courts to over-rely on AI reasoning and ignore what it is not detecting. All of this creates problems because if the AI is biased then the evidence it produces is also biased. This is why there need to be safeguards in place before AI evidence is allowed in court (NIST, 2022).

Fixing this issue is not simple and there are several things that must happen. Forensic investigations must require explainable AI. These systems must provide the logic and reasoning behind its decisions, which would help address the “black box” AI problem. Explainable AI systems need to disclose what algorithms they use, what data the model was trained on, and where the model has limitations. That way courts can actually evaluate whether the findings are reliable. Experts could testify about why the AI model flagged something and defense attorneys would have something concrete to challenge. Independent human verification must confirm AI findings to ensure the evidence is accurate. Legal professionals should receive mandatory

training on AI capabilities, limitations, and how to evaluate AI-generated evidence. With these safeguards in place, AI can still be used in forensic investigations, but in a way that maintains integrity of the evidence that holds up in court.

Ensuring these safeguards would require continuous follow up and review. AI tools are going to keep changing and becoming more advanced, so the policies and procedures around them must keep up with these changes. There should be independent oversight of error rates, bias, and court acceptance of AI evidence. Developers would be responsible for meeting updated standards and retraining AI models to ensure data remains admissible.

All three frameworks are similar when it comes to AI forensic investigations. They all care about transparency, fairness, and accountability, but how they get there is different. The Christian worldview is based on the Ten Commandments and emphasizes moral duty to God and to your neighbor. The ACM and IEEE perspectives emphasize professional standards and benefiting society. These are more about professional and practical responsibility, but they line up with the Christian values closely. I believe our legal system is based on Christian values, and the ACM and IEEE standards end up aligning with those values. Between the three of them, they cover the ethical ground needed to make sure AI evidence is used the right way.

Allowing ethical use of AI forensic evidence presents challenges in court. The “black box” problem keeps coming up because courts need to know how AI got to its answer before they will accept it. The solutions I discussed all come back to the same issue which is ensuring AI evidence can be explained, verified, and held to a standard before it is allowed in court.

Alternative solutions included allowing juries to decide how much weight to give AI evidence. This was ultimately rejected as juries lack the technical knowledge to evaluate AI

findings this way. Another consideration was focusing on standardizing AI evidence collection procedures rather than requiring AI to explain its findings. The issue with this is having procedures does not always guarantee accurate analysis. Finally, just leaving the standards to individual courts was rejected because it creates unfairness and inconsistency in the justice system. At the end of the day, AI can be a powerful tool in forensic investigations, but if it is not held to ethical standards, people are going to lose trust in the system.

References

ACM. (2018). *ACM code of ethics and professional conduct*. https://www.acm.org/code-of-ethics

IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems. (2019). *Ethically aligned design: A vision for prioritizing human well-being with autonomous and intelligent systems* (1st ed.). IEEE. https://standards.ieee.org/industry-connections/ec/autonomous-systems/

National Institute of Standards and Technology. (2022). *Towards a standard for identifying and managing bias in artificial intelligence* (NIST Special Publication 1270). U.S. Department of Commerce. https://doi.org/10.6028/NIST.SP.1270

OpenAI. (2024). *ChatGPT* (December 2024 version) [Large language model]. https://chat.openai.com/chat

Solanke, A. A. (2022). Explainable digital forensics AI: Towards mitigating distrust in AI-based digital forensics analysis using interpretable models. *Forensic Science International: Digital Investigation*, *42*, Article 301403. https://doi.org/10.1016/j.fsidi.2022.301403

Stoykova, R. (2024). A new right to procedural accuracy of automated decision-making? *Computer Law & Security Review*, *55*, Article 106040. https://doi.org/10.1016/j.clsr.2024.106040